

Universidade de Coimbra
Faculdade de Ciências e Tecnologia
Departamento de Engenharia Electrotécnica e de Computadores

Sistemas de Tempo Real - 2026/2027
Practical Assignment Nº1
Scheduling, Priorities Assignment, and Measurement of Computation Times

Introduction

The work to be performed consists on the implementation of an application that makes use of the POSIX routines, with the goal of studying the scheduling by priorities assignment and the measurement of the computation times. There are three functions, `f1(class,group)`, `f2(class,group)` and `f3(class,group)` assigned to three distinct periodic tasks. These functions have as input parameters two integer numbers that identify the class ("turma", 1, 2 or 3, corresponding to PL1, PL2 or PL3) and the group. The corresponding code is in the files `func.h` and `func.o`, and must be compiled with the application developed.

This work should be performed in POSIX [1], [2] with Real-Time Extension [1], [3]. The *kernels* of recent distributions already (and also) largely implement the real-time part of the POSIX *standard*. If, and only if, some part of the work may not be strictly performed with the POSIX functionalities, then for that part of the work it can also be used functions of the "Official Linux" [4], [5] (for example, functions of the Linux library).

All the *software* in this practical assignment should be developed/configured to run in only one processor. Even if the work is developed in a system with only one processor, the *software* should be developed in a manner that it executes in only one processor if later the *software* is compiled and executed in a computer with multiple processors.

The work should be preferably developed in a 64 bits Linux architecture.

Description

The items of the work are as follows:

1. Using the POSIX functionalities that you may consider necessary, measure the (maximum) computation time of each of the functions. The given functions are expected to run within a very short time variation (small execution ripple).

2. Using the results of point 1 verify if the system is schedulable, for two priority orderings: RMPO (rate monotonic priority ordering) and its inverse. Present in the report the calculations performed. The three tasks should be periodically activated with periods of 0.1 [s], 0.2 [s] and 0.3 [s]. Use Gantt charts and the Audsley method. Present in the report all the calculations and charts done.

3. Write the code of an application that, during about 6 seconds, maintains the three tasks jointly running using RMPO. Tasks 1 to 3 should be periodically activated with periods of 0.1 [s], 0.2 [s] and 0.3 [s], respectively. The application should show if the tasks have met the deadlines. To do so, prior to the activation of each task and after its execution, a log table should be filled with those time instants, being printed at the end of the application. In the end, the application should also show the largest response times for each task and the response time jitter.

4. Alternative A: Change the code developed in Item 3 in order to change the priorities (between RMPO and the inverse) during the execution of the tasks at instant $t = 1.95$ [s] and at $t = 3.95$ [s]. Comment the obtained results.
   Alternative B: Change the code developed in Item 3 in order to run in any number of processors

(CPU cores). Try also to invert the priorities (to invert RMPO priorities). Comment the obtained results, namely the largest response times for each task and if they met the deadlines.

5. Develop a source code `func2.c` of a module `func2.o` that imitates the functioning of the module `func.o`. Test the module `func2.o` using it jointly with the created application. For this purpose, you should in particular repeat the work of Item 3 above, using `func2.o` in substitution of `func.o`. Comment the obtained results.

6. Assigning equal priorities to the tasks, test the application developed in Item 3 with the Round Robin scheduling method. Comment the results, namely the largest response times for each task and the response time jitter.

POSIX and Linux Documentation

Resources on the Internet

• **POSIX Specification** – `http://www.opengroup.org/onlinepubs/9699919799/`

• **POSIX and its Real-Time Extension** –
Consult and study the references, investigating what are the parts of POSIX which are most important for the POSIX part of the work. Some topics to study: *pthreads*, their attributes (e.g. priorities) and scheduling options [1], functionalities for measurement and management of time. Consult [6], including the example in C. Consult the **POSIX specification** [1], including the following parts:
▷ System Interfaces ▷ General Information ▷ Realtime;
▷ Topic ▷ Realtime;
▷ System Interfaces ▷ General Information ▷ Threads.

• **Linux *Manpages*** – `http://www.kernel.org/doc/man-pages/`
The pages of the Linux manual may have relevance to the development of the work. Section "2: System calls" may have particular relevance.

Hints

Use of pthreads
Some relevant functions:
`pthread_create()`
`pthread_join()`
`pthread_exit()`
`mlockall()`
Compilation flags: `-D_REENTRANT -lpthread`

Compilation with the POSIX real-time extension
Compilation flags: `-lrt`

RTAI Documentation

Internet Resources

• **RTAI official web page** – `https://www.rtai.org/`
In "Documentation ▷ Reference Documents ▷ RTAI User Manual" there is the official manual. Chapters 5 and 6 are the most important. It is the best available reference. This manual may contain some typos.
In the upper right corner of the page there is an engine to make searches in the mailing list.
May contain outdated parts.

Report and Material to Deliver

A report should be delivered to the professor in `PDF` format. The report should contain the following information in the first page: name of the course ("disciplina"), title and number of the practical assignment, names of the students, number of the class ("turma"), number of the group. It should be submitted through the Nónio system (`https://inforestudante.uc.pt/`) area of the "Sistemas de Tempo Real" course in a single file in `zip` format that should contain all the relevant files that have been involved in the realisation of the practical assignment. The name of the submitted `zip` file should be "`tXgYrZ-str27.zip`", where "`X`", "`Y`" e "`Z`" are characters that represent the numbers of the class ("turma"), group, and practical assignment, respectively.

References

[1] The IEEE and The Open Group. [POSIX Specification] The Open Group Base Specifications Issue 7; IEEE Std 1003.1^{TM}-2008, 2008. [Online]. Available: `http://www.opengroup.org/onlinepubs/9699919799/` .

[2] Wikipedia. POSIX. [Online]. Available: `http://en.wikipedia.org/wiki/POSIX` .

[3] The Realtime Extension: A White Paper from the X/Open Base Working Group, 1997. [Online]. Available: `http://www.unix.org/version2/whatsnew/realtime.html` .

[4] The Linux Man-Pages Project. The Linux Man-Pages Project. [Online]. Available: `http://www.kernel.org/doc/man-pages/` .

[5] Linux Kernel Documentation. Linux Kernel Documentation Index. [Online]. Available: `http://www.kernel.org/doc/` .

[6] [Linux] RT PREEMPT HOWTO. [Online]. Available: `http://rt.wiki.kernel.org/index.php/RT_PREEMPT_HOWTO` .